CHEMISTRY

Max.Marks:100 Time Allowed: 3hrs

Theory: 70 Marks Practicals: 30 Marks

UNIT-1: SOME BASIC CONCEPTS OF CHEMISTRY 07 Marks

General Introduction: Importance of studying chemistry, Historical approach to particulate nature of matter, Laws of Chemical combination, Dalton's Atomic Theory, Concept of elements, atoms & molecules. Atomic and molecular masses, Mole concept and molar mass, percentage composition, empirical and molecular formula; chemical reactions, stoichiometry and calculation based on stoichiometry.

Unit-II: STRUCTURE OF ATOM 09 Marks

Discovery of electron, proton and neutron, atomic number, isotopes and isobars. Thomson's model and its limitations, Rutherford's model and its limitations. Bohr's model & its limitations, concept of shells and sub-shells. Dual nature of matter and light, de-Broglie's relationship. Heisenberg's uncertainty principle, concept of orbitals, quantum numbers, shapes of s, p and d- orbitals. Rules for filling electrons in orbitals- Aufbau's principle, Pauli's exclusion principle and Hund's rule. Electronic configuration of atoms, stability of half-filled and completely filled orbitals.

Unit-III: CLASSIFICATION OF ELEMENTS AND PERIODICITY IN PROPERTIES 06 Marks

Significance of classification, brief history of the development of periodic table. Modern periodic law and the present form of the periodic table, periodic trends in properties of elements: atomic radii, ionic radii, inert gas radii, ionization enthalpy, electron gain enthalpy, electronegativity, valency.

Unit-IV: CHEMICAL BONDING AND MOLECULAR STRUCTURE 07 Marks

Valence electrons, Ionic bond, covalent bond, bond parameters, Lewis structure, polar character of covalent bond, valence bond theory, resonance, geometry of covalent molecules, VSEPR theory, concept of hybridization involving s, p and d-orbitals and shapes of some simple molecules, molecular orbital theory of homonuclear molecules (Qualitative idea only), hydrogen bonding.

Unit-V: THERMODYNAMICS 09 Marks

Concepts of system, types of systems, surrounding, work, heat; energy intensive and extensive properties, state functions. First Law of Thermodynamics, internal energy, enthalpy, heat capacity, specific heat, molar heat capacity, measurement of ΔE and ΔH, Hess's law of constant heat summation, enthalpy of bond dissociation, combustion; formation, atomization, sublimation, phase transition ionization and dilution.

Introduction of entropy as a state function, free energy change for spontaneous and non-spontaneous process and criteria for equilibrium.

Unit-VI: EQUILIBRIUM 07 Marks

Equilibrium in physical and chemical processes, dynamic nature of equilibrium, law of mass action, equilibrium constant, factors affecting equilibrium: Le-Chatelier's principle: ionic equilibrium-ionization of acids and bases, strong and weak electrolytes, degree of ionization, Concept of pH. Hydrolysis of salts (elementary idea), buffer solutions, solubility product, common ion effect (with suitable examples).

Unit-VII: REDOX REACTIONS 04 Marks

Concept of oxidation and reduction, redox reactions, oxidation number, balancing of chemical equations in redox reactions, applications of redox reactions.

Unit-VIII: ORGANIC CHEMISTRY-SOME BASIC PRINCIPLES AND TECHNIQUES

11 Marks

General introduction to organic chemistry, methods of purification, qualitative and quantitative analysis, classification and IUPAC nomenclature of organic compounds.

Electronic displacement in a covalent bond: inductive effect, electromeric effect, resonance and hyper-conjugation. Homolytic and heterolytic fission of a covalent bond, free radicals, electrophiles, nucleophiles, carbocations and carbanions. Types of organic reactions.

Unit-IX: HYDROCARBONS 10 Marks

Classification of hydrocarbons

Alkanes: Nomenclature, isomerism, conformations (ethane only), Methods of preparation, physical properties. Chemical reactions including free radical mechanism of halogenation, combustion and pyrolysis

Alkenes: Nomenclature, structure of double bond (ethene), geometrical isomerism, methods of preparation, physical properties, chemical reactions- addition of hydrogen. halogen, water, hydrogen halides (Markownikov's addition and peroxide effect). ozonolysis, oxidation, mechanism of electrophilic addition.

Alkynes: Nomenclature, structure of triple bond (ethyne), physical properties, methods of preparation, chemical reactions: acidic character of alkynes, addition reaction of hydrogen, halogens, hydrogen halides and water.

Aromatic hydrocarbons: Introduction, IUPAC nomenclature, Benzene resonance, aromaticity, chemical properties, mechanism of electrophilic substitution-nitration, sulphonation, halogenations Friedel Craft's alkylation and acylation, directive influence of functional group in mono substituted benzene.

Scheme of Assessment (Theory)

Question paper contains four sections namely

1. Section–A contains10 (Objective Type Question/Multiple Choice Questions) of1mark each.1x10= 10 marks
2. Section-B contains 9 Very Short Answer Type Questions of 2 marks each to be answered in 20 to 30 words.2 x 9 = 18 marks
3. Section–C contains 9 Short Answer type questions of 3 marks each to be answered in100 to 150 words. 3 x 9 = 27 marks
4. Section-D contains 3 Long Answer type questions of 5 marks each to be answered in150 to 200 words. 5 x 3 = 15 marks

Note: The paper setter shall incorporate Higher Order Thinking Skills (HOTS)questions apart from knowledge, memory and application based questions.

PRACTICALS
Time: 03 Hours

Maximum Marks: 30
External: 20 Marks/ Internal: 10 Marks

1. CONTENT BASED EXPERIMENTS 06 Marks

A) Organic Preparations:

i) Preparation of acetylene and study of its acidic character.

ii) Preparation of Acetanilide

iii) Preparation of p-Nitro acetanilide

B) Characterization and Purification of Chemical Substance:

i)Determination of melting point of an organic compound (below 100°C)

ii) Determination of boiling point of an organic liquid.

iii) Crystallization involving impure sample of any one of the following: Alum, Copper sulfate, Benzoic acid.

C) Experiments Related to pH Change

Any one of the following experiments:

i) Determination of pH of some solutions obtained from juices and solutions of known and varied concentrations of acids, bases and salts using pH paper/ universal indicator

ii) Comparing the pH of solutions of strong and weak acid of same concentration

iii) Study the pH change in the titration of a strong acid with a strong base using universal indicator

iv) Study of pH change by common-ion effect in case of weak acids and weak bases

D) Chemical Equilibrium:

One of the following experiments

i) Study the shift in equilibrium between ferric ions and thio cyanate ions by increasing/decreasing the concentration of either ion

ii) Study the shift in equilibrium between $[Co(H_2O)_6]$ and Cl^- ions by changing the concentration of either ions.

2. Quantitative Estimation: 08 Marks

i) Setting of a chemical balance and preparation of a standard solution of oxalic acid

ii) Determination of strength of a given sodium hydroxide solution by titrating it against a standard solution of oxalic acid.

iii) Preparation of standard solution of sodium carbonate.

iv) Determination of strength of given solution of dilute hydrochloric acid by titrating it against a standard solution of sodium carbonate.

3. Qualitative Analysis 08 Marks

Determination of one cation and one anion in a given salt (insoluble salts to be excluded):

Cations: Pb^{2+}, Cu^{2+}, As^{3+} Al^{3+}, Fe^{3+}, Mn^{2+},.Zn^{2+},Ni^{2+}, Co^{2+}, Ca^{2+}, Sr^{2+}, Ba^{2+}, Mg^{2+}, NH_4^+,

Anions:CO_3^{2-}, S^{2-}, SO_3^{2-}, SO_4^{2-},NO_2^-, NO_3^-, Cl^-, Br^-, PO_4^{3-},$C_2O_4^{2-}$, CH_3COO^-

4. PROJECT 04 Marks

Scientific investigation involving laboratory testing and collecting information from other sources.

- Study of the methods of purification of water.
- Checking the bacterial contamination in drinking water by testing sulphide ion.
- Testing the hardness, presence of Iron, Fluoride, Chloride, etc., depending upon the regional variation in drinking water and study of causes of presence of these ions above permissible limit (if any).
- Analysis of fruit and vegetable juices for their acidity.
- Preparation of a sample of soap from available oils (Groundnut/Coconut oil).

- Determination of the rate of evaporation of different liquids.
- Study of the effect of acids and bases on the tensile strength of fibers.
- Compare the contents of tannic/ caffeine in various samples of tea and hence their flavor.

5. Class Record and Viva-Voce 04 Marks

Note: Collaboration to seek from nearby Institutions with regard to the performing of practicals/project work.

BOOKS PRESCRIBED:

1. *A textbook of Chemistry for class XI published by NCERT, New Delhi.*